SHAP-σ: A Label-Free Rashomon Pipeline for Multivariate Time-Series Anomaly Detection

Ugur Dar[1*], Jeriek Van den Abeele[2], Mustafa Cavus[1], Sara Malacarne[2]
[1]Eskisehir Technical University, Department of Statistics, Eskisehir, Türkiye
[2]Telenor Research and Innovation, Oslo, Norway
*ugurdarr@gmail.com

***Abstract*—We address unsupervised anomaly detection in multivariate cellular network KPI streams, where labels are unavailable for training, model selection, and thresholding. We propose SHAP-σ, a label-free pipeline that self-tunes a heterogeneous set of anomaly detectors and selects a compact Rashomon set—a collection of structurally different models exhibiting comparable performance. Within this framework, the highest-quality detector is used as a reference *anchor* to flag anomalies at the timestamp level. This initial flag is then validated across the remaining models in the Rashomon set by evaluating the cross-detector standard deviation of their SHAP feature attributions. Finally, an attribution-evidence filtering mechanism refines these global alarms, retaining only the specific KPIs that show substantial cross-detector variance, thereby turning global outputs into localized, KPI-level alarms.**

On the TELCO benchmark, SHAP-σ improves over both the best singleton remaining after surrogate-fidelity filtering and the best overall self-tuned singleton under a controlled comparison. Compared with the DC-VAE and C-MTAD-GAT results under the same per-KPI aggregation, it achieves more than a twofold relative improvement in macro F_1. The results show that most of the performance gain comes from label-free self-tuning and Rashomon-set selection, while the attribution filtering mainly contributes precision and interpretability by attaching explicit KPI-level evidence to each retained alarm. The pipeline is therefore label-free and more transparent than a black-box ensemble vote.

***Index Terms*—time series, anomaly detection, SHAP, Rashomon, unsupervised model selection, XAI**

I. Introduction

Unsupervised anomaly detection in multivariate time series underpins fault diagnosis in domains as diverse as cellular networks, server fleets, and industrial telemetry. In cellular networks, operators monitor many time-indexed KPIs, such as traffic, availability, throughput, and error-related counters. Yet, a persistent obstacle hinders deployment: without ground-truth labels, the practitioner cannot reliably decide which detector to trust. This problem is particularly difficult in operational telemetry, where non-stationarity, changing traffic patterns, and heterogeneous KPI behavior can make a detector effective on some streams but unreliable on others. In the absence of labels, model selection remains difficult because many internal validation scores do not reliably predict which anomaly detector will perform best [3].

Unsupervised multivariate time series anomaly detection is commonly framed as learning normal temporal behavior and flagging deviations through reconstruction, forecasting, density, or distance-based scores. Classical deep approaches include sequence autoencoders and encoder–decoder models [**?**], variational models such as OmniAnomaly [**?**], and GAN-based detectors such as MAD-GAN [**?**]. More recent methods use graph attention or Transformer-style architectures to capture long-range temporal structure and cross-variable dependencies, including MTAD-GAT [**?**], TranAD [16], Anomaly Transformer [17], and DCdetector [18]. Telecom-specific work includes DC-VAE [1] for mobile-network KPI anomaly detection [**?**] and C-MTAD-GAT [2] for context-aware graph-attention-based monitoring at operator scale. These methods improve the modeling of temporal and cross-KPI structure, but practical deployment still depends on label-free thresholding, score aggregation, and model selection.

This paper focuses on the last of these issues: how to use several plausible unsupervised detectors when labels are not available to choose among them. This setting naturally leads to predictive multiplicity: different detectors may appear equally reasonable on the same data, while flagging different timestamps or associating alarms with different KPIs. This is closely related to the Rashomon effect, where many models can be similarly plausible while making different errors or emphasizing different features [4]–[6]. Choosing a single detector in this setting can therefore create unjustified trust in one explanation or decision [**?**]. Recent work shows that Rashomon sets can also be used constructively, by aggregating model behavior to quantify uncertainty and obtain more robust explanations [**?**]. We follow this perspective in the unsupervised setting: instead of treating disagreement among plausible detectors as noise, we use it as a signal for anomaly localization.

We propose SHAP-σ, a label-free pipeline for anomaly detection and KPI-level localization in multivariate cellular network streams. The method first self-tunes a heterogeneous pool of unsupervised detectors and selects a small label-free Rashomon set using quality and diversity criteria. Different detector families encode different assumptions, such as distance, density, histogram, projection, or kernel structure, and may therefore attribute the same anomaly to different KPIs. SHAP-σ exploits this variation as follows: the highest-quality

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The codes and data for reproducibility are given at: `https://github.com/ugurdar/shap_sigma_paper`.

detector in the selected set acts as an *anchor*, raising the initial timestamp-level anomaly flag, and the flag is then retained only for KPIs whose surrogate-based attributions [12], [13] show high cross-detector variation.

Prior work on label-free model selection has mainly focused on ranking or choosing detectors from their global anomaly scores. Excess-Mass and Mass-Volume curves [7] provide geometry-based criteria; MetaOD [8] and ELECT [9] use meta-learning from historical labelled datasets; and robust Borda aggregation combines multiple imperfect internal metrics for time-series anomaly detection [10]. However, large-scale empirical studies show that no internal criterion universally dominates [3]. These approaches address label-free detector selection, but they do not use feature-attribution information to decide which KPI should be associated with a detected anomaly.

SHAP [12], [13] is widely used as a post-hoc explanation method, typically after a single model has already been selected. The closest related work is Levy et al. [14], who use SHAP attributions for ensemble member *selection* in supervised tabular anomaly detection. In contrast, SHAP-σ uses cross-detector SHAP standard deviation as an attribution-based *gate* in an unsupervised time-series setting. This also connects to outlier ensemble theory [19], [20] and classical diversity measures [21]; however, standard diversity measures such as Q-statistics and disagreement require labels. Following [20], we therefore use Spearman rank correlation between training anomaly scores as a label-free proxy for detector diversity.

The contributions of this paper are as follows:

- **Label-free self-tuning** of a heterogeneous pool of $K{=}10$ unsupervised detectors (PCA, k-NN, LOF, GMM, HBOS, IF, OCSVM, ECOD, COPOD, and a PyTorch AutoEncoder), using only the training data and no anomaly labels.
- **Label-free Rashomon-set selection** using a joint quality–diversity score $s_{ij} = \frac{1}{2}(q_i{+}q_j)(1{-}\rho_{ij})$, where q_d is a Mass–Volume quality proxy and ρ_{ij} is the Spearman correlation between detector scores.
- **A SHAP-σ attribution gate** that retains anchor-generated anomaly flags only for KPIs whose surrogate-based attributions show high cross-detector variation, yielding a transparent KPI-level localization signal.
- **Empirical validation** on the TELCO cellular benchmark, showing that the label-free pipeline improves over the best self-tuned singleton detector under a controlled comparison, while reporting higher per-KPI macro F_1 than the published DC-VAE and C-MTAD-GAT results, which are used only as contextual references.

The remainder of the paper is organized as follows. Section II formalizes joint pair scoring and the SHAP-σ filter. Section III describes datasets and protocol. Section IV reports empirical results. Section V concludes.

II. Methodology

SHAP-σ is a label-free pipeline for multivariate time-series anomaly detection with KPI-level localization. A single *anchor* detector decides *when* a timestamp is anomalous, while the selected Rashomon set decides *which* KPIs support that alarm through the cross-detector variation of SHAP attributions. No labels are used for training, tuning, ranking, set construction, or thresholding; labels serve only for final evaluation.

A. Notation

Let $X_{\mathrm{tr}} \in \mathbb{R}^{N_{\mathrm{tr}} \times p}$, $X_{\mathrm{v}} \in \mathbb{R}^{N_{\mathrm{v}} \times p}$, and $X_{\mathrm{te}} \in \mathbb{R}^{N_{\mathrm{te}} \times p}$ be the training, validation, and test matrices, and $\mathcal{D} = \{d_1, \ldots, d_K\}$ the detector pool; each d produces a score $s_d(x) \in \mathbb{R}$s. Detectors are fit on X_{tr}; X_{v} is a label-free calibration split for every threshold; X_{te} is used only for final evaluation. We select a Rashomon set $\mathcal{R} \subseteq \mathcal{D}$ whose highest-quality member is the anchor a; the remaining members contribute attribution variation, not votes.

B. Detector Pool and Label-Free Self-Tuning

The pool mixes empirical-CDF, histogram, distance, density, projection, isolation, kernel, and neural detectors: ECOD [22], COPOD [23], HBOS [24], k-NN [25], LOF [26], PCA [27], GMM [19], Isolation Forest [28], OCSVM [29], and a PyTorch AutoEncoder (the first eight via PyOD [11]). This heterogeneity is what makes cross-detector attribution differences informative. Inputs are standardized, but each detector's raw scores are used as-is downstream, since the later steps are either rank-based or applied per detector.

Off-the-shelf defaults can fail, so each parameterized detector is self-tuned on X_{tr} only: PCA keeps the components explaining 95% of variance; k-NN and LOF take a k-distance elbow over $k \in \{1, \ldots, 50\}$; GMM picks its component count by BIC over $\{2, \ldots, 8\}$; HBOS bins by Sturges' rule; Isolation Forest uses the smallest tree count in $\{50, \ldots, 500\}$ beyond which the training-score standard deviation changes by under 2%; the AutoEncoder is fixed to a $[64, 32]$ architecture; OCSVM uses an RBF kernel with $\nu{=}0.05$; ECOD and COPOD are parameter-free. All choices depend only on X_{tr} and use no labels.

C. Stage 1 – Label-Free Quality via a Mass–Volume Proxy

Following the Mass–Volume principle [7], a useful detector concentrates its high scores on a small region rather than over a uniform background. We fit a LightGBM surrogate $\hat{s}_d$ on $(X_{\mathrm{tr}}, s_d(X_{\mathrm{tr}}))$ and estimate the background tail mass

$$A_d = \Pr_{X_{\mathrm{u}}}\big[\hat{s}_d(X_{\mathrm{u}}) > Q_{0.9}\big(s_d(X_{\mathrm{tr}})\big)\big], \tag{1}$$

where $Q_{0.9}$ is the 0.9-quantile of training scores; a lower A_d means more selective scoring. A min–max inversion yields the quality

$$q_d = \frac{\max_e A_e - A_d}{\max_e A_e - \min_e A_e} \in [0, 1]. \tag{2}$$

Eq. (1) is estimated by Monte Carlo over three random low-dimensional subspaces with $2{,}000$ uniform samples each. We use this single-level proxy rather than full EM/MV integration for light, reproducible computation [3].

D. Stage 2 – Joint Rashomon-Set Selection

Diversity is the Spearman rank correlation $\rho_{ij} = \mathrm{Spearman}(s_i(X_{\mathrm{tr}}), s_j(X_{\mathrm{tr}}))$, with $1-\rho_{ij}$ as the pairwise measure. The *Mixed* strategy balances quality and diversity; for a pair it selects

$$(i^*, j^*) = \arg \max_{1 \le i < j \le K} \frac{q_i + q_j}{2} (1 - \rho_{ij}). \tag{3}$$

More generally, for a target size K_R the Mixed strategy selects the subset

$$\mathcal{R} = \arg \max_{|S|=K_R} \bar{q}(S)\, \bar{\delta}(S), \tag{4}$$

where $\bar{q}(S)$ and $\bar{\delta}(S)$ average the quality q_d and the pairwise diversity $1-\rho_{ij}$ over S. The set size K_R is a hyperparameter; selecting it automatically and label-free is a separate model-selection problem that we leave to future work, and we instead report performance across K_R (Sec. IV). Unless stated otherwise, the K_R=2 pair is the default operating point. The anchor is the highest-quality selected detector, $a = \arg\max_{d \in \mathcal{R}} q_d$.

E. Stage 3 – Surrogate SHAP Attribution

To place detectors in a common explanation space, each $d \in \mathcal{R}$ is approximated by a LightGBM surrogate g_d [32] on $(X_{\mathrm{tr}}, s_d(X_{\mathrm{tr}}))$, from which we take as exact TreeSHAP attributions $\Phi_d^{\mathrm{v}}, \Phi_d^{\mathrm{te}}$ [12], [13] normalized by their validation scale,

$$\tilde{\Phi}_d^{\bullet} = \frac{\Phi_d^{\bullet}}{\mathrm{std}(\Phi_d^{\mathrm{v}}) + \varepsilon}, \qquad \varepsilon = 10^{-12}. \tag{5}$$

$$R_d^2 = R^2!\big(s_d(X_{\mathrm{v}}), g_d(X_{\mathrm{v}})\big), \quad \mathcal{D}_{\mathrm{f}} = d \in \mathcal{D} : R_d^2 \ge 0.75$$

Surrogates are used only for explanation: the anchor flag still comes from the original score s_a, so surrogate fidelity affects only the gate and cannot by itself raise an alarm. We nonetheless measure each surrogate's fidelity by the validation R^2 between s_d and g_d, and discard detectors with low fidelity from the candidate pool, since their attributions are unreliable; on TELCO this removes LOF and KNN. Selection (Stage 2) and the σ gate then operate on the remaining detectors.

F. Stage 4 – The SHAP-σ Attribution-Evidence Gate

For each timestamp i and KPI j we measure the cross-detector attribution variation

$$\sigma_{ij} = \mathrm{std}_{d \in \mathcal{R}}\big(\tilde{\Phi}_{d,ij}\big), \tag{6}$$

where a large value means the plausible detectors disagree on KPI j's contribution. A per-KPI threshold estimated on the validation set,

$$\tau_j = Q_q\big(\{\sigma_{ij}^{\mathrm{v}}\}_{i=1}^{N_{\mathrm{v}}}\big), \qquad q = 0.75, \tag{7}$$

gives the label-free gate

$$\phi_{ij} = \mathbf{1}\big[\sigma_{ij}^{\mathrm{te}} > \tau_j\big]. \tag{8}$$

Algorithm 1 SHAP-σ (label-free)

Require: $X_{\mathrm{tr}}, X_{\mathrm{v}}, X_{\mathrm{te}}$; pool $\mathcal{D}$ (K=10); quantile q; threshold κ.
Ensure: per-KPI predictions $\hat{Y}$ (optional point-level $\hat{y}$).
1: Fit every $d \in \mathcal{D}$ on X_{tr}; score all splits.
2: Compute quality q_d via Eqs. (1)–(2).
3: Fit a surrogate g_d per detector; drop those with $R^2_{\mathrm{val}} < 0.75$ (low fidelity).
4: $\mathcal{R} \leftarrow$ Mixed subset of size K_R (Eq. (4)); anchor $a \leftarrow \arg\max_{d \in \mathcal{R}} q_d$.
5: **for all** $d \in \mathcal{R}$ **do**
6: TreeSHAP $\Phi_d^{\mathrm{v}}, \Phi_d^{\mathrm{te}}$; normalize (Eq. (5)).
7: **end for**
8: Compute σ_{ij}, τ_j, ϕ_{ij} (Eqs. (6)–(8)).
9: Compute anchor flag $f_i^{(a)}$ (Eq. (9)).
10: $\hat{Y}_{ij} \leftarrow f_i^{(a)} \wedge \phi_{ij}$

G. Stage 5 – Anchor Flag and σ-Gated Output

The anchor raises a κ-sigma flag from its training-score statistics (μ_a, σ_a),

$$f_i^{(a)} = \mathbf{1}\big[s_a(x_i^{\mathrm{te}}) > \mu_a + \kappa\sigma_a\big], \qquad \kappa = 3, \tag{9}$$

and a flag is kept for KPI j only where attribution evidence agrees,

$$\hat{Y}_{ij} = f_i^{(a)} \wedge \phi_{ij}. \tag{10}$$

Thus the anchor decides *when* and SHAP-σ decides *which* KPI. TELCO has per-KPI labels, so Eq. (10) is evaluated by per-KPI macro F_1; with only point-level labels it collapses to

$$\hat{y}_i = f_i^{(a)} \wedge \Big(\bigvee_{j=1}^{p} \phi_{ij}\Big). \tag{11}$$

III. Experiments

A. Dataset

TELCO dataset [1] from an operational mobile cellular network spanning January–July 2021, with p=12 KPIs sampled every five minutes and ground truth labelled by Network Operation Center experts. The official test phase contains 26,496 timestamps per KPI; summed over the 12 streams there are 3,080 anomaly-labelled points.

B. Protocol

We split the training data into 80% training and 20% validation; detectors are fit on training only. Hyperparameters are self-tuned label-free on the training split (Sec. II-B). Quality q_d uses three random subspaces with 2,000 uniform background samples per subspace. The SHAP-σ quantile is fixed at q=0.75 and the anchor κ-sigma at κ=3.

TABLE I
PER-KPI MACRO METRICS OF THE 10 SELF-TUNED SINGLETONS.

Detector	F_1	P	R	Tune
LOF	**0.419**	0.460	**0.452**	k = elbow
ECOD	0.401	**0.488**	0.396	param-free
KNN	0.115	0.209	0.089	k = elbow
GMM	0.112	0.478	0.066	BIC
AutoEncoder	0.098	0.130	0.089	[64, 32], 50 ep
OCSVM	0.092	0.105	0.099	ν=0.05
IF	0.058	0.047	0.092	path-len. plateau
COPOD	0.056	0.052	0.072	param-free
PCA	0.042	0.035	0.067	95% var
HBOS	0.033	0.349	0.017	Sturges

C. Evaluation Metric

We report precision, recall and F_1 under three aggregation schemes: *Macro*, *Micro* and *Union*. In the Macro setting, each metric is computed independently on each of the 12 KPI binary label streams and then averaged uniformly across KPIs; this corresponds to a per-KPI macro score. In the Micro setting, predictions and labels are pooled across all KPIs before computing precision, recall and F_1. In the Union setting, KPI-level binary predictions are combined by a logical OR at each timestamp, so that a timestamp is counted as anomalous if any KPI is predicted anomalous.

For the deep baselines DC-VAE and C-MTAD-GAT, we quote the per-KPI macro precision, recall and F_1 reported in their papers under the same Macro aggregation protocol; we do not re-train these models.

IV. RESULTS AND DISCUSSION

Table I reports the per-KPI macro F_1, precision, and recall of every self-tuned singleton detector on TELCO when its κ-sigma flag is broadcast across the 12 KPIs. Self-tuned LOF leads at F_1=0.419, narrowly above ECOD; all remaining detectors stay below 0.12. The strongest singletons LOF and ECOD differ qualitatively: LOF is distance-based with 0.452 recall, while ECOD's per-feature CDF favours precision, making them natural pair candidates downstream.

We first discard detectors whose surrogate fidelity is low; on TELCO this removes LOF and KNN, leaving an eight-detector pool with reliable attributions. Figure 2 shows all K_R=2 pairs under the closed-form joint pair score; its naive maximiser is LOF+ECOD, but LOF is exactly the detector the fidelity filter removes. After filtering, the results in Table II keeps F_1 in a narrow 0.42–0.44 band across K_R, and its best operating point is the K_R=2 pair ECOD+HBOS at F_1=0.442 with the highest precision; ECOD is the anchor. Because choosing K_R automatically and label-free is a separate problem, we report the whole sweep and take the K_R=2 pair as the default operating point; adding further detectors does not raise F_1 and slightly lowers precision.

Table III dissects the default operating point per KPI. SHAP-σ achieves F_1>0.7 on TS3 and TS5, F_1>0.55 on TS4 and TS6, and about 0.52 on TS7 and TS8; it fails only on the TS11 and TS12 with $<$ 25 ground-truth anomalies each. On these two streams the method emits only false positives : at such low prevalence the per-KPI validation quantile τ_j admits cross-detector variation that does not coincide with the few true anomalies; in deployment they would need a prevalence-aware threshold or exclusion, but we keep them in the macro average for transparency. The pair's main strength is precision: the macro precision is 0.647, and per-KPI precision exceeds 0.78 on seven KPIs.

TABLE II
F_1 UNDER MIXED JOINT-SUBSET SELECTION

K_R	Selected members (added on each row)	F_1	P	R
2	ECOD+HBOS	**0.442**	**0.647**	0.374
3	+OCSVM	0.435	0.590	0.384
4	+GMM	0.431	0.586	0.381
5	+COPOD	0.428	0.579	0.382
6	+PCA	0.431	0.582	0.383
7	+IForest	0.424	0.571	0.382
8	+AE	0.428	0.575	**0.383**

TABLE III
PER-KPI BREAKDOWN OF THE DEFAULT OPERATING POINT: SHAP-σ WITH K_R = 2 (ECOD+HBOS)

KPI	#true	#pred	TP	FP	FN	F_1	P	R
TS1	412	122	103	19	309	0.386	0.844	0.250
TS2	501	127	100	27	401	0.318	0.787	0.200
TS3	138	121	105	16	33	**0.811**	0.868	**0.761**
TS4	179	113	99	14	80	0.678	0.876	0.553
TS5	106	101	79	22	27	0.763	0.782	0.745
TS6	107	139	83	56	24	0.675	0.597	0.776
TS7	269	179	117	62	152	0.522	0.654	0.435
TS8	267	113	102	11	165	0.537	**0.903**	0.382
TS9	564	127	102	25	462	0.295	0.803	0.181
TS10	501	160	104	56	397	0.315	0.650	0.208
TS11	21	34	0	34	21	0.000	0.000	0.000
TS12	15	54	0	54	15	0.000	0.000	0.000
macro	3,080	1,390				0.442	0.647	0.374
micro	3,080	1,390	994	396	2,086	0.445	0.715	0.323
union	1,232	181	120	61	1,112	0.170	0.663	0.097

Table IV compares SHAP-σ against the deep baselines on the per-KPI macro F_1. The per-KPI macro F_1 of SHAP-σ is clearly higher than the DC-VAE and the C-MTAD-GAT.

However, these baselines are quoted rather than re-trained here, and their reported scores are not obtained under the same self-tuning, thresholding, and per-KPI macro-F1 evaluation protocol used in our experiments. We therefore treat these numbers only as contextual references, not as a controlled comparison. The controlled comparison is against the best self-tuned singleton: SHAP-σ exceeds the best singleton in the filtered pool by +0.041, and also the best overall singleton as given in Table I. Its contribution is to refine detector outputs by localising alarms to the responsible KPI, rather than only improving the global anomaly score.

Most of the improvements comes from the label-free self-tuning and the fidelity-filtered selection rather than from the σ gate's raw F_1: a single self-tuned LOF already reaches 0.419, and ECOD reaches 0.401. The σ gate's distinctive value is therefore not a higher F_1 but (i) *precision*: it raises macro precision to 0.647 and per-KPI precision up to 0.90, suppressing false alarms; and (ii) *interpretability*: unlike a black-box ensemble vote, it attaches an explicit per-KPI

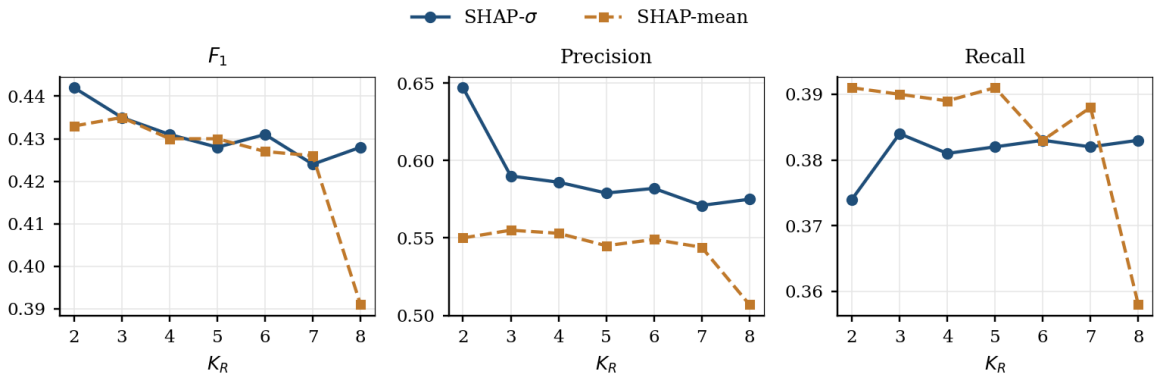


Fig. 1. SHAP-σ vs SHAP-mean across K_R. SHAP-σ gives consistently higher precision.

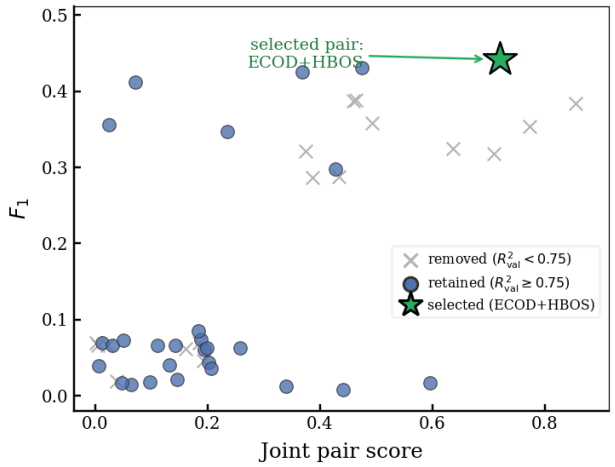


Fig. 2. Joint pair score vs. F_1 for all K_R=2 pairs on TELCO. The naive maximiser includes LOF, which the surrogate-fidelity filter removes; after filtering, the selected pair is ECOD+HBOS.

TABLE IV
PERFORMANCE SUMMARY: SHAP-σ WITH $K_R = 2$ (ECOD+HBOS) VS. DC-VAE AND C-MTAD-GAT.

	Macro			Micro			Union		
Method	F_1	P	R	F_1	P	R	F_1	P	R
DC-VAE	0.134	0.220	0.096	0.084	0.127	0.063	0.107	0.103	0.110
C-MTAD-GAT	0.160	0.319	0.107	0.084	0.133	0.062	0.116	0.117	**0.115**
SHAP-σ	**0.442**	**0.647**	**0.374**	**0.445**	**0.715**	**0.323**	**0.170**	**0.663**	0.097

attribution to every surviving flag, identifying *which* KPI supports each alarm. Code and per-KPI outputs are released for reproducibility.

The gate can aggregate the per-detector attributions in different ways. Figure 1 contrasts our cross-detector *disagreement* signal SHAP-σ, with the natural alternative of cross-detector *magnitude* SHAP-mean, under the identical anchor, threshold, and K_R sweep. The two reach comparable F_1, but SHAP-σ is consistently more precise at every K_R and stays stable as the set grows, whereas SHAP-mean trades precision for a recall and degrades for large K_R . Since our target is precision-first KPI localization – suppressing false alarms rather than maximizing recall – we adopt SHAP-σ: *disagreement* among plausible detectors about a KPI's contribution is a cleaner localization signal than its average attribution magnitude.

V. CONCLUSIONS

We presented SHAP-σ, a label-free pipeline for multivariate cellular network anomaly detection that promotes the cross-detector standard deviation of SHAP attributions from a post-hoc explanation into a first-class attribution-evidence signal. The methodological contributions are label-free self-tuning of a Rashomon set, a closed-form Rashomon pair score combining a label-free Mass–Volume quality proxy with Spearman score-decorrelation, and a per-KPI σ gate that filters anchor flags by the cross-detector spread of explanations.

On the TELCO cellular benchmark, after discarding low-fidelity detectors, the label-free quality–diversity selection reaches the best operating point with the ECOD+HBOS pair and ECOD as the anchor. It improves over the best singleton remaining in the filtered pool and also outperforms the best overall singleton. Choosing the set size K_R is a separate model-selection problem left to future work; performance remains within a narrow range across the tested set sizes. We are explicit that this gain comes chiefly from label-free self-tuning and selection: the σ gate's contribution is not primarily raw F_1, but higher precision and – unlike a black-box vote – explicit per-KPI attribution evidence for every retained flag.

The DC-VAE and C-MTAD-GAT baselines report lower per-KPI macro F_1 than SHAP-σ, but they are quoted from prior work rather than retrained under our self-tuning, thresholding, and evaluation protocol. We therefore treat them as contextual references rather than as a controlled comparison. Overall, the pipeline remains label-free, transparent, reproducible, and explicitly per-KPI.

By construction the gate is recall-monotone: it can only suppress anchor flags, so SHAP-σ is a *precision-first* filter, not a recall booster. We regard this as appropriate for the target setting – in operational KPI monitoring the dominant cost is alert fatigue from false positives rather than marginal

recall – but it does cap recall at the anchor's, and pairing the gate with a higher-recall anchor is left to future work. The attribution step relies on the surrogate tracking each detector; we therefore discard low-fidelity detectors rather than letting their unreliable attributions enter the gate, though model-native attributions would be preferable where available. The disjunctive feature-wise gate is most informative on per-feature labels and degenerates under OR-aggregation for point-level evaluation, which is deferred to a journal extension. We also plan to learn label-free from synthetic anomaly injection [8], [10], to compare against deep label-free baselines such as MTGFlow [33], and to extend SHAP-σ to cross-window variance for streaming deployment.

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